

Tutorial-4

2.

An Email Service Provider wants to classify emails as Spam or not Spam Using logistic Regression.
Data from 10 emails has been collected.

Input-features:

No. of Suspicious words.

No. of External links

Presence of attachments (0=No, 1=Yes)

x_1	x_2	x_3	y
5	2	1	1
1	0	0	0
3	1	0	1
0	0	0	0
6	3	1	1

1. Identify Independent & Dependent Variables.

Independent Variables.

x_1 = Suspicious words

x_2 = External links

x_3 = Attachments

$x = (x_1, x_2, x_3)$

Dependent Variables

y = Spam

(1 = Spam,

0 = Not Spam)

2. Write Logistic Regression Model. Eqn.

Logistic Regression.

$$Z = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$$

Sigmoid fn.

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$w_1 = w_2 = w_3 = 0$$

$$Z = 5(0) + 2(0) + 1(1) + 1 = 1 \quad \hat{y} = 1$$

$$Z_2 = 1 \quad \hat{y} = 0.73 \quad y = 1$$

$$Z_3 = 1 \quad \hat{y} = 0.73 \quad y = 1$$

$$Z_4 = 1 \quad \hat{y} = 0.73 \quad y = 1$$

$$Z_5 = 1 \quad \hat{y} = 0.73 \quad y = 1$$

$$\text{Accuracy} = \frac{3}{5} = 60\%$$

$$J = \frac{1}{N} \sum_{i=1}^N (y_i \log \hat{y}_i + (1-y_i) \log (1-\hat{y}_i))$$

$$= \frac{1}{5} [(1) \log(0.73) + (1-1) \log(1-0.73) + \\ (0) \log(0.73) + (1-0) \log(1-0.73) + \\ (1) \log(0.73) + (1-1) \log(1-0.73) + \\ (0) \log(0.73) + (1-0) \log(1-0.73) + \\ (1) \log(0.73) + (1-1) \log(1-0.73)]$$

$$0.73 \rightarrow 1-0.73 \rightarrow 0.27$$

Update weights & bias

$$\boxed{w_0^{\text{new}} = w_0^{\text{old}} + \eta \frac{1}{N} \sum_{i=1}^n (y - \hat{y}_i)}$$

$$= 1 + 1 \left(\frac{1}{5} \right) [(1 - 0.73) + (0 - 0.73) + (1 - 0.73) + (0 - 0.73) - 1(1 - 0.73)]$$

$$= 1 + \left[\frac{-0.65}{5} \right]$$

$$\approx 1 - 0.13 \rightarrow 0.87$$

$$w_1^{\text{new}} = w_1^{\text{old}} + \eta \left[\frac{1}{N} \sum_{i=1}^n (y - \hat{y}_i) x_i \right]$$

$$w_1 = 0 + 1 \times \frac{1}{5} [(0.27)(5) + (-0.73 \times 1) + (0.27 \times 3) + (-0.73 \times 0) + (0.27 \times 6)] = 0.302$$

$$w_2 = 0 + 1 \times \frac{1}{5} [(0.27)(2) + (-0.73 \times 0) + (0.27 \times 1) + (-0.73 \times 0) + (0.27 \times 3)] = 0.02$$

$$w_3 = 0 + 1 \times \frac{1}{5} [(0.27)(1) + (-0.73)(0) + (0.27)(0) + (-0.73)(0) + (0.27)(1)]$$

$$= 0.108$$

An Enc
or Sp

Input - f

Target

x₁

with w

Sample 1:

if y [A
yE